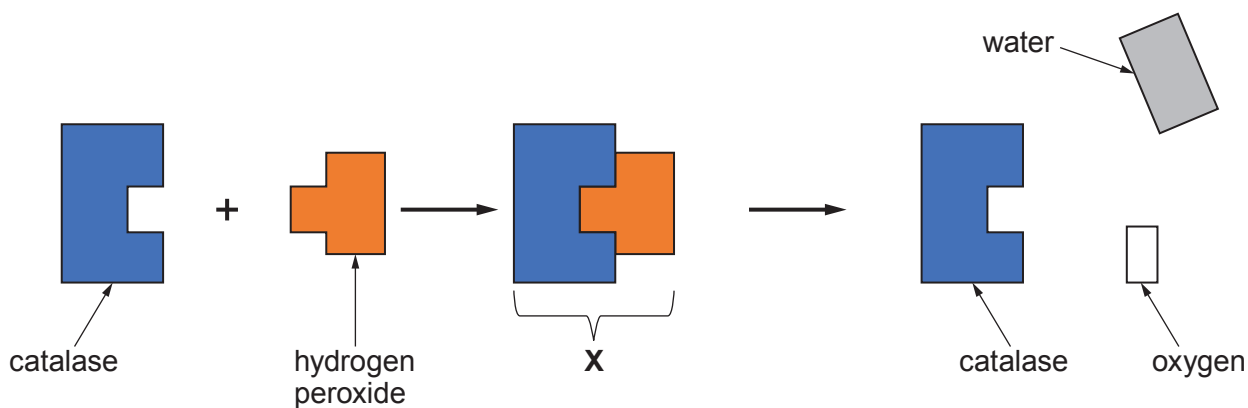


9. Catalase is an enzyme found in most cells. Catalase speeds up the breakdown of hydrogen peroxide to form water and oxygen.
Image 9.1 represents the breakdown of hydrogen peroxide by the action of catalase.

Image 9.1

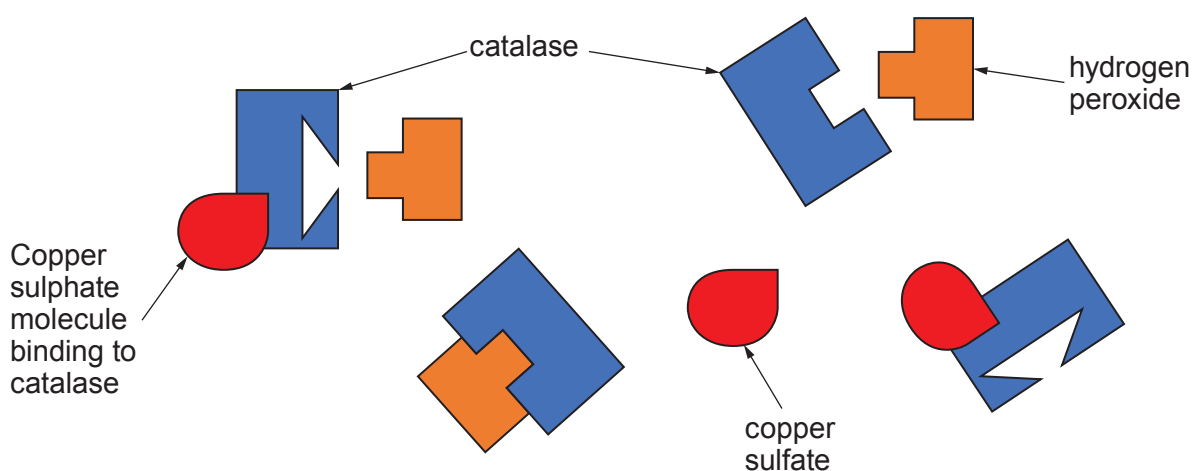


- (a) (i) State the name of the structure labelled **X** in **Image 9.1**. [1]

- (ii) State the name of the model of enzyme action shown in **Image 9.1**. [1]

- (b) **Image 9.2** represents the effect of copper sulfate on catalase. If a copper sulfate molecule binds to catalase it prevents it from working.

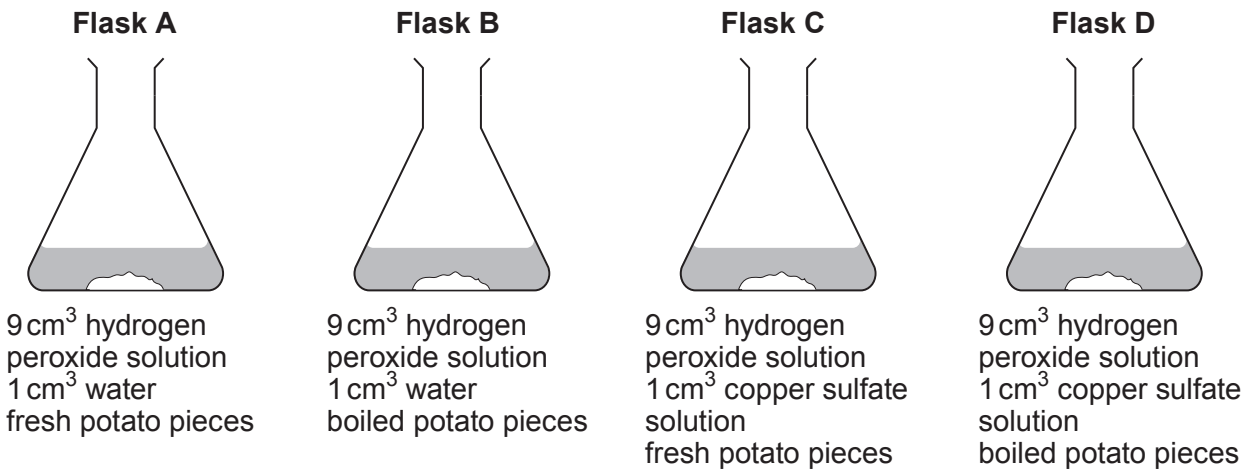
Image 9.2



Students investigated the effect of copper sulfate on catalase activity in potatoes.

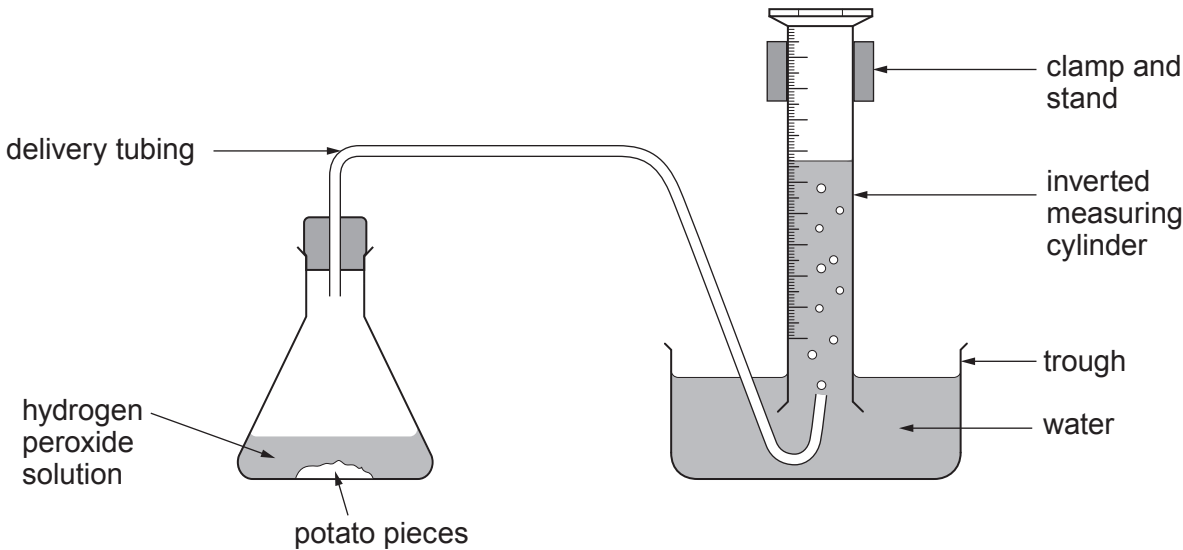
The students set up the following four flasks. as shown in **Image 9.3**.

Image 9.3



The students measured the volume of gas produced from each flask using the apparatus shown in **Image 9.4**.

Image 9.4



The results of the investigation are shown in **Table 9.5**.

Table 9.5

Time (minutes)	Volume of gas present in the cylinder (cm ³)			
	Flask A	Flask B	Flask C	Flask D
0	0	0	0	0
5	12	0	3	0
10	20	0	5	0
15	28	0	7	0
20	34	0	10	0
25	38	0	12	0
30	38	0	14	0

(i) Describe **and** explain the results for flask **A**. [4]

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(ii) The rate of gas production in flask **A** between 10 and 15 minutes is 1.6 cm³/min.
Calculate the rate of gas production between 10 and 15 minutes in flask **C**. [2]

Rate of gas production = cm³/minute



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(iii) Use the information in **Image 9.2** to explain why less gas is produced in flask **C** than in flask **A**. [3]

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(c) (i) State the purpose of flasks **B** and **D** in **Image 9.3**. [1]

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(ii) State **two** variables that should have been controlled in this investigation to make it a fair test. [2]

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(iii) Describe how the students could improve the accuracy of their investigation. [1]

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15

END OF PAPER

